

# Christian Carrico

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## PROJECTS

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### AFSIM Reinforcement Learning Environment

- Designed and delivered solutions for a Python Reinforcement Learning environment that integrates AFSIM and graph based observations to optimize blue force outcomes
- Developed code to procedurally generate sufficiently random AFSIM scenarios, providing a reinforcement learning agent with enough experiences to enhance generalization across the entire problem space
- Built Regex- and Jinja2-powered scripts to parse hundreds of AFSIM model files into a Python interface, enabling better scenario generation and a richer feature space
- Optimized Docker and Docker Compose configurations, reducing environment development image size by 60%, dramatically reducing build times and accelerating deployments to a classified network

### Modeling & Simulation Framework Trade Study

- Directed research and assessment of multiple M&S frameworks, coordinating tasks with government and contractor personnel
- Developed objective evaluation criteria and formalized use cases to determine the most suitable framework
- Compiled findings into a comprehensive report and presented key insights to leadership for decision-making

### Workflow Planning Web Application

- Led the development of a COCOM exercise planning tool by using Agile methodologies to integrate feedback from Warfighters into tasks for the next sprint
- Enhanced codebase quality by refactoring C# controllers and standardizing JavaScript web views, improving maintainability and simplifying feature delivery

## EXPERIENCE

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### U.S. Army DEVCOM Aviation & Missile Center

Colorado Springs, CO

*Chief Engineer*

*Mar 2025 – Present*

- Led teams of government, FFRDC, and contractor personnel by organizing and delegating tasks to deliver software capabilities to the Warfighter
- Briefed software architecture diagrams to senior agency leadership, shaping high-level technical strategy and decisions
- Designed and initiated development of a scalable software platform to operationalize AI/ML research in support of Missile Defense

### U.S. Army DEVCOM Aviation & Missile Center

Colorado Springs, CO

*Software Engineer*

*Apr 2018 – Mar 2025*

- Employed Amazon Web Services to deploy critical infrastructure such as CI/CD pipelines and collaboration tools to support various government efforts
- Leveraged technologies such as the Atlassian tool suite and GitLab to capture customer requirements, document design decisions, and accelerate the development process
- Partnered with small businesses on SBIR/STTR contracts, overseeing progress and deliverables while providing technical guidance to maximize government value
- Utilized SysML and Cameo to develop system models, supporting early-stage architecture design, analysis, and CONOPs
- Mentored student interns and new employees by delegating meaningful tasks that foster skill development and professional growth
- Evaluated research proposals for technical and commercial merit while ensuring alignment with Agency priorities

### Electrical & Computer Engineering College

Fort Collins, CO

*Student Ambassador*

*Aug 2012 – Aug 2014*

- Engaged prospective students during recruitment events to promote interest in the college's programs and drive enrollment growth

## EDUCATION

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### Colorado State University

Fort Collins, CO

*B.S. in Computer Engineering*

*Aug 2012 – Dec 2017*

*B.S. in Mathematics, Concentration in Information Security*

*Aug 2012 – Dec 2017*